1. Clinical Question & Core Hypothesis

Clinical Question

PSMA-PET lesion burden and PSA kinetics do not always align—some patients have low/negative 68Ga-PSMA-11 imaging yet rapidly rising PSA (high PSA velocity) and early recurrence. Why does imaging-defined low visible disease sometimes fail to predict durable control when biochemical dynamics indicate active disease?

Working Hypothesis

Primary expanded hypothesis: In men with localized prostate cancer who experience biochemical recurrence after radical local therapy, a discordant PSMA–PSA profile defined by low 68Ga-PSMA-11 PET lesion burden (low-volume or PSMA-silent disease) but rapid PSA kinetics (high PSA velocity or rising PSA) represents a biologically aggressive, PSMA-low subclone with micrometastatic dissemination that escapes PET detection. This discordant phenotype will be associated with shorter progression-free interval (PFI) after salvage or systemic-intensification strategies and will identify patients who derive greater absolute and relative benefit from early systemic therapy (e.g., docetaxel-containing regimens or intensified hormonal therapy) compared with patients whose PSMA and PSA signals are concordant. Secondary hypotheses: (a) the discordant group will have a higher circulating tumor DNA burden and/or PSMA-low molecular signatures, indicating micrometastatic disease; (b) quantitative PSMA-PET metrics (e.g., SUVmax, PSMA volume) will inversely correlate with PSA velocity; (c) a combined PSMA–PSA risk score will improve prediction of PFI/FFS beyond either modality alone; (d) early systemic therapy will attenuate the adverse impact of discordance on progression-free outcomes. Trial-relevant endpoints include primary PFI and secondary endpoints such as FFS, metastasis-free survival, and time to PSA progression.

Element	Value
Imaging feature	68Ga-PSMA-11 PET lesion burden
Anchor gene(s)	PSA velocity, serum PSA value
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PSA velocity: PSA velocity → Recurrence/Progression → FFS.

serum PSA value: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Docetaxel	Supporting modifier from knowledge graph

Secondary	STAMPEDE	Supporting modifier from knowledge graph
Secondary	Prostate RT	Supporting modifier from knowledge graph
Secondary	Radical Prostatectomy	Supporting modifier from knowledge graph
Secondary	Stages	Supporting modifier from knowledge graph
Secondary	Failure-Free Survival (FFS)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: The interaction between the imaging feature and anchor gene predicts clinical outcome.

Prediction 2: Spatial analysis will confirm the mechanistic link at the tissue level.

Prediction 3: Functional perturbation will establish causality.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Docetaxel: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- STAMPEDE: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.